

DriftSense: Privacy-Preserving Intention-Aware Prediction and Model-Assisted Reflection for Browser-Based Digital Drift

[AUTHOR NAME], [Department / University], [Country]

Most digital wellbeing tools rely on one of three ideas: block a domain, limit screen time, or show the user a usage dashboard. The difficulty is that none of these approaches explains why a person opened a website. The same site may support a focused task, an intentional break, or an unplanned period of browsing. This paper presents DriftSense, a privacy-preserving Chrome extension and local machine-learning pipeline for studying browser-based digital drift. We define drift as a mismatch, reported by the participant, between the intention stated before a browsing session and the outcome reported afterward. The extension records configured-domain metadata, aggregate interaction counts, and short activity-window summaries. It does not collect page content, screenshots, messages, or keystroke values. We describe a two-phase field study with approximately 20 adult Chrome users. Participants first complete a 10-day observation phase in which they declare an intention and reflect after each session. We then compare time, domain, intention-only, activity-only, and intention-plus-activity prediction approaches and freeze one three-minute model and threshold. During a seven-day intervention phase, eligible elevated-risk sessions are randomized equally to a reflective prompt or a silent control. Prediction is assessed using chronological and participant-aware validation, while the prompt is assessed using only randomized eligible sessions. Data collection has not yet taken place; accordingly, this paper reports the system and planned methodology without fabricating results. DriftSense concerns self-reported intention alignment, not attention, emotion, addiction, or mental health diagnosis.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; *Ubiquitous and mobile computing systems and tools*; • **Computing methodologies** → *Machine learning approaches*.

Additional Key Words and Phrases: digital wellbeing, browser extension, intention alignment, digital drift, self-report, privacy-preserving sensing, session-level prediction, reflective check-ins

1 Introduction

Opening a website that is often described as distracting does not necessarily mean that a person has become distracted. Someone may open YouTube for a tutorial, visit Reddit to solve a technical problem, check a message on LinkedIn, or follow a particular event on a news site. Those same websites can also lead to unplanned passive browsing. This ambiguity matters because a domain name or a clock cannot, by itself, distinguish purposeful use from digital drift.

Many existing tools define undesirable use through screen time, website category, or a fixed blocking rule [3, 13–15]. These controls may help, but they can also interrupt a legitimate visit or overlook a short session that has already moved away from its purpose. More recent work has therefore paid greater attention to agency, reflection, friction, and intention. For example, *one sec* inserts a brief pause before app access [9]; Purpose Mode changes attention-capturing interface patterns while keeping the site available [11]; and PauseNow treats digital wellbeing as alignment between a user’s intention and subsequent behavior [17].

We call the session-level mismatch between an initial purpose and a later self-report *browser-based digital drift*. The term is intentionally narrow. DriftSense does not infer an internal mental state, a clinical condition, or objective attention. Instead, the participant states an intention before browsing and later reports whether the session remained consistent with it. The research question is whether that self-reported outcome can be anticipated from a small set of browser activity signals.

DriftSense consists of a local-first Chrome Manifest V3 extension and a reproducible machine-learning pipeline. Before entering a configured domain, the participant selects a neutral intention: work or study, learning or a tutorial,

Author’s Contact Information: [Author Name], [Department / University], [City], [Country], [Email].

specific information, communication or community, a planned entertainment break, open-ended browsing, or an accidental opening. These choices avoid treating work as automatically aligned or leisure as automatically problematic. While the session is active, the extension records duration, aggregate clicks and scrolls, idle time, focus changes, and short activity-window summaries. It then asks whether the visit matched the original intention. In Phase 2, a local three-minute prediction determines whether the session is eligible for randomization: one half of eligible sessions receive a brief reflective prompt and the other half receive no mid-session interface. Access to the website is preserved in both cases.

The study asks two questions:

RQ1: Can declared browsing intention combined with lightweight browser activity signals predict self-reported session-level drift better than time-based or domain-based approaches?

RQ2: Among sessions classified as elevated risk after three minutes, does a model-assisted reflective prompt reduce subsequent self-reported drift compared with a randomized silent control?

The intended contribution is not another dashboard, activity logger, or blocker. It is a focused implementation and evaluation of intention-aware session prediction: privacy-preserving signals, an intention-grounded label, explicit weak baselines, leakage-safe early prediction, and a randomized test of the resulting prompt. Phase 1 supplies model-development data; it is not treated as a causal control for Phase 2.

2 Related Work

2.1 Digital Self-Control Interventions

Blocking, time limits, usage monitoring, reminders, friction, and interface changes are recurring strategies in digital self-control tools [3, 13, 14]. Systematic reviews report some encouraging short-term findings, but they also show a literature dominated by brief studies, inconsistent outcomes, convenience samples, and limited use of control groups [14, 15]. Time spent is widely reported, although it says little about the purpose of a particular visit. Purpose Mode offers a useful contrast: it retains access while allowing attention-capturing interface patterns to be suppressed, and its results suggest that goal alignment can be more informative than website context alone [11]. DriftSense therefore treats time and domain as comparison baselines rather than definitions of drift.

2.2 Intention and Reflection

Several systems create a moment in which users can reconsider what they are doing. *one sec* places friction and an explicit continue-or-dismiss choice before app access [9]. PauseNow makes original intention central to its conceptual framework, although it does not report a deployment or outcome evaluation [17]. StayFocused and MindShift explore reflective or generated interventions; their studies also reveal mixed behavioral outcomes and concerns about repetition, incorrect assumptions, and the collection of richer personal context [12, 16]. WellScreen takes a personal-informatics approach by comparing estimated and actual smartphone use and prompting daily reflection, but its two-week study did not include a control condition [2]. DriftSense draws on the reflective principle while using a brief nonblocking prompt and avoiding mental-state data.

2.3 Session-Level Modeling and Research Gap

Work on regretful technology use shows why duration is an incomplete measure: regret varies with the activity and the user's intention [7, 10]. In a seven-day sensing study with 21 participants, perceived intention–use mismatch was more

Table 1. Closest systems informing the DriftSense research gap.

Work	Timing and mechanism	Role of intention	Evaluation	Remaining gap
Purpose Mode [11]	During-use interface modification	Goal alignment measured in EMA	Two-week field deployment	No session prediction; fixed phase order
<i>one sec</i> [9]	Friction before app access	Opening reason elicited around nudge	Observational field study and online experiment	No within-session prediction or field control
PauseNow [17]	Proposed recognition and nudging	Original intention is central	Conceptual framework	No deployed outcome evaluation
<i>Before You Scroll Again</i> [1]	Prediction before social-media use	Mismatch reported after use	Seven-day sensing study; within-person and held-out tests	Regret label; no intervention test
WellScreen [2]	Daily post-use reflection	Estimated versus actual use	Two-week mixed-methods deployment	No session classifier or control
DriftSense (planned)	Three-minute model-assisted reflection	Declared before and revisited after use	Ten-day collection plus randomized seven-day intervention	Browser-only pilot with short follow-up

strongly related to regret than duration. The reported phone-context CatBoost model reached AUC 0.732 within person but only 0.536 when one participant was held out [1]. That difference makes it important to report known-user and unseen-user performance separately. It also motivates a smaller privacy boundary: DriftSense uses hostname-level context and aggregate browser activity rather than page content, screenshots, or wearable physiology [4, 5].

Table 1 summarizes the closest comparison points. None of these evaluated systems brings together a declared browser intention, content-free early activity signals, a post-session alignment label, weak prediction baselines, and a randomized model-triggered prompt.

The resulting contribution is deliberately limited. We ask whether intention and lightweight activity improve session-level prediction over time, domain, intention-only, and activity-only alternatives, and whether a prompt based on that model performs better than a randomized silent control. A short pilot cannot demonstrate durable behavior change or broad population-level generalization.

3 System Design

3.1 Evidence-Derived Design Requirements

DriftSense is intended for voluntary use in a research study: participants install the extension, choose the domains to monitor, and control the local data that they later export. The reviewed work leads to five requirements:

- (1) **Assess alignment rather than assuming harm.** Time and domain remain useful baselines, but the outcome must depend on what the participant intended and later reported. Prior findings suggest that goal alignment and intention–use mismatch can be more informative than duration [1, 11].
- (2) **Keep the intervention light and preserve choice.** The check-in must not block the website. This design follows evidence that friction and an explicit opt-out can interrupt an automatic action, while repetitive or inflexible prompts can become irritating [9, 12].
- (3) **Collect less and keep it local.** DriftSense uses aggregate event counts and hostname-level context. It excludes page content, screenshots, messages, and key values, and lets participants inspect, export, or delete local records.

- (4) **Use consistent measurement.** Intention and post-session questions remain the same across both phases and both randomized assignments. The mid-session prompt is the only randomized interface difference.
- (5) **Leave an auditable trail.** Exports must support the required baselines, early prediction at one, three, and five minutes, participant-aware validation, a frozen deployment model, and verification of eligibility and randomization.

Figure 1 presents the planned extension and local learning pipeline.

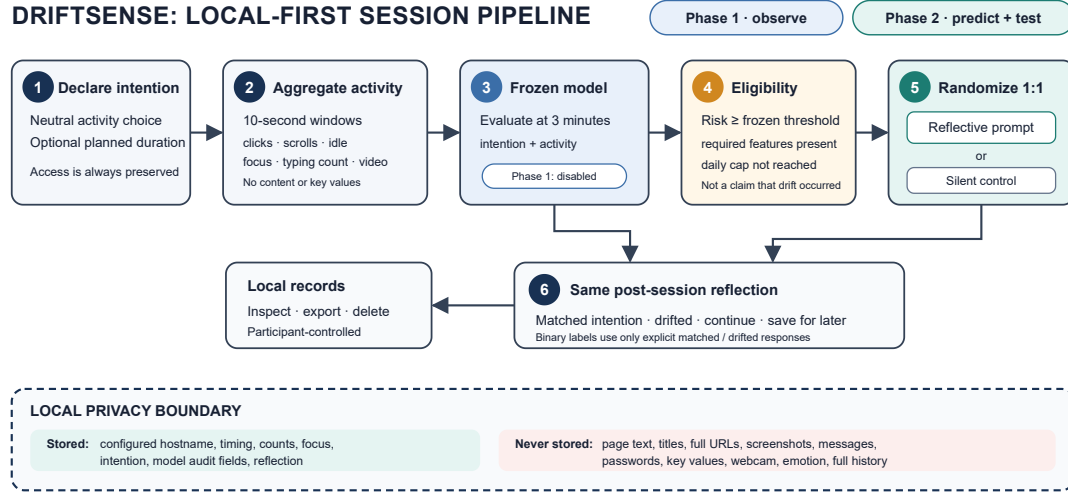


Fig. 1. DriftSense architecture and intervention decision flow. Prediction and randomization are disabled during the 10-day collection phase and enabled with a frozen model during the seven-day intervention phase.

3.2 Chrome Extension Architecture

The collection prototype is implemented as a Chrome Manifest V3 extension using React and TypeScript. A background service worker manages sessions, content scripts count aggregate activity on monitored domains, and the popup and options pages expose status, domain settings, and privacy controls. A local dashboard supports summaries and export. The intervention version adds a fixed model evaluated after three minutes, a frozen eligibility threshold, session-level 1:1 randomization, and the reflective prompt. Remote inference is not required.

3.3 Session Flow, Sensing, and Labeling

Participants choose the domains they want DriftSense to study; activity outside those domains is ignored. When a configured domain opens, the extension asks *Why are you opening this?* The seven response categories cover work/study, learning/tutorial, specific information, communication/community, planned entertainment/break, open-ended browsing, and accidental opening. The categories describe purpose without judging it.

A session begins when a configured domain opens or receives focus. It ends when the tab closes, leaves that domain, reaches the timeout rule, or completes reflection. The main session record contains anonymous participant and session identifiers, start time, hostname, intention and intended duration, total duration, aggregate click/scroll/keyboard-activity counts, idle time, focus losses, and the reflection outcome. Linked 10-second windows contain only offset, aggregate

event counts, idle/focus state, and accessible video-playback state. A separate audit log records model version, threshold decision, eligibility, assignment, prompt delivery and response, and suppression reason. The schema documentation provides the exact field names.

Every session uses the same post-session question:

Did this visit match your original intention?

Only explicit *Yes* and *No, I drifted* responses define the binary target:

$$y = \begin{cases} 0, & \text{if answer is Yes} \\ 1, & \text{if answer is No, I drifted} \end{cases} \quad (1)$$

Continue intentionally, *Save for later*, and missing responses remain separate and are not recoded as non-drift. During Phase 2, an eligible session assigned to the prompt condition shows *Still here for your original reason?* with *Continue intentionally* and *Finish now*. A silent-control session shows nothing at that point. The probability is never displayed as a fact, and access remains available. Dashboarding supports participant control and inspection; it is not presented as the research contribution.

4 Drift Prediction and Model-Assisted Reflection

4.1 Prediction Task

The modeling problem is binary and session based:

Given a declared intention, hostname-level context, and lightweight browser activity observed so far, predict whether the participant will later report that the session drifted.

No, I drifted is the positive class and *Yes* is the negative class. Non-binary and missing responses are kept outside the primary classification task. The output is a session-level probability. In Phase 2, crossing the frozen threshold establishes eligibility for prompt-versus-silent randomization; it is not interpreted as attention detection.

4.2 Feature Groups

The feature groups include intention, domain category, time and day, elapsed duration, event and focus-loss counts, tab switching, idle/active/video ratios, and activity rates. Windows are truncated at one, three, and five minutes. The intervention uses the three-minute version, while the other cutoffs and the full session are evaluated offline. Features observed after a cutoff, including final duration, are never used for an earlier prediction.

Figure 2 separates the prediction evaluation from the randomized prompt evaluation.

4.3 Baselines

Four comparisons are evaluated before the combined model:

- (1) **Time threshold.** Predict drift when duration exceeds a threshold selected on validation data.
- (2) **Domain rule.** Estimate domain-level drift tendency using training data only, then freeze that rule for validation or testing.
- (3) **Intention-only logistic regression.** Use the declared intention and basic context without activity features.
- (4) **Activity-only logistic regression.** Use aggregate activity without the declared intention.

TWO PHASES, TWO DISTINCT EVALUATION QUESTIONS

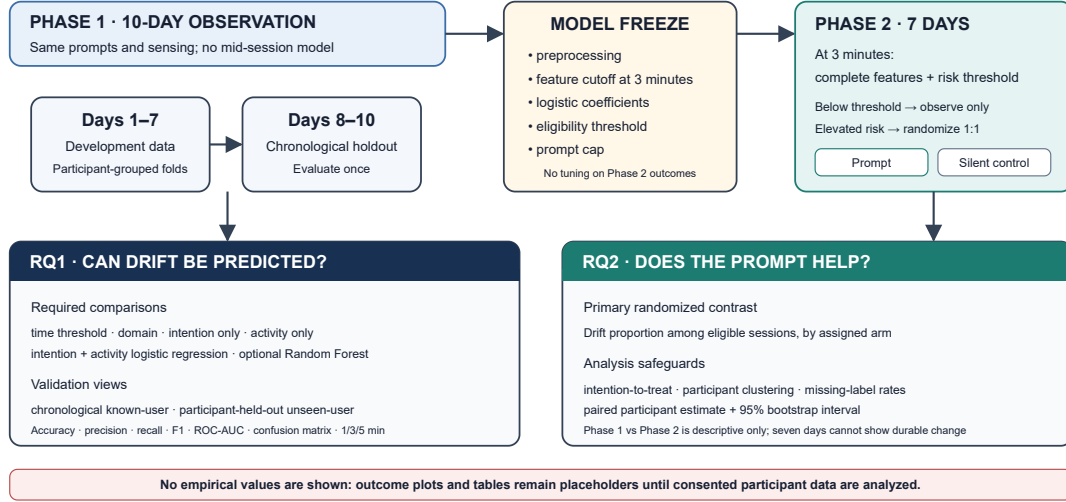


Fig. 2. Planned study and evaluation pipeline. RQ1 uses Phase 1 to evaluate prediction; RQ2 uses randomized eligible Phase 2 sessions to evaluate the prompt. The overall Phase 1–Phase 2 change is not the causal intervention contrast.

Together, these comparisons test whether combining intention and activity adds information beyond common digital wellbeing heuristics.

4.4 Main Models

The primary model is intention-plus-activity logistic regression. It is interpretable, suitable for a modest dataset, and straightforward to deploy locally as fixed preprocessing, coefficients, an intercept, and a threshold. Random Forest is an optional nonlinear comparison if the available sessions and class balance support it. CatBoost or XGBoost may be included as secondary offline checks [6, 8]. Deep sequence models are not planned because 15–20 participants are unlikely to provide enough independent data to justify them.

4.5 Early Prediction

Offline results will compare one-, three-, and five-minute cutoffs with full-session performance. Three minutes is selected in advance for deployment: it is early enough to support reflection but avoids interrupting the user immediately after entry. The model family, preprocessing, threshold, and prompt cap are fixed before Phase 2. A threshold crossing creates eligibility for randomization; it neither forces a prompt nor proves that drift occurred.

5 Methodology

5.1 Study Design

The planned pilot follows one cohort for 17 study days. Phase 1 is a 10-day observation period with the same pre-session intention and post-session reflection prompts used throughout the study, but no model-assisted mid-session prompt. After model development, Phase 2 runs for seven days with the frozen model. Sessions that remain active at

three minutes and cross the frozen threshold are randomized equally to a reflective prompt or a silent control. This session-level randomized contrast is the primary evidence for RQ2. The overall difference between Phase 1 and Phase 2 will be descriptive because the fixed order, novelty, and changes over time prevent a causal interpretation.

The study targets 20 completers, with 22–24 recruits to allow for attrition. Operational feasibility targets are 500 complete binary-labeled Phase 1 sessions overall, approximately 20 labeled Phase 1 sessions per participant where browsing frequency permits, and 100 randomized elevated-risk Phase 2 sessions. These targets are not a formal power calculation. If the study falls well short of them, the paper will be framed as a feasibility study and the intervention estimate as exploratory.

5.2 Participants and Recruitment

Participants must be adults, use Chrome or another Chromium-based browser on a personal computer, regularly visit at least two domains they are comfortable monitoring, and be able to install the research extension. Recruitment may rely on a university convenience sample of students and knowledge workers. No one is recruited or screened on the basis of addiction, ADHD, mental health, or another clinical category. Reporting will include recruitment source, relevant demographics, attrition, usable-session counts, and missing labels without claiming that the sample is representative.

5.3 Consent, Privacy, and Data Handling

Before installation, participants receive consent information and a field-level description of the data. Consent covers configured hostnames, intention and intended duration, session timing, aggregate click/scroll/keyboard-activity counts without key values, idle time, focus loss, activity windows, and post-session responses. In Phase 2 it also covers model version and estimate, random assignment, prompt delivery, and prompt response. DriftSense does not collect page text, passwords, screenshots, private messages, full browsing history, source code, keystroke values, webcam data, face identity, or emotion.

Participants use anonymous codes such as P01. They can pause monitoring, change domains, inspect or export records, and delete individual sessions or all local data. Exports are stored outside the repository in a private access-controlled study directory. Only consented research staff can access the combined data. Publications will use aggregates or appropriately de-identified excerpts rather than raw participant files. Ethics approval or institutional exemption must be obtained before recruitment.

5.4 Phase 1: Ten-Day Collection

During onboarding, each participant selects domains from work/learning and mixed-use contexts. For every monitored session, DriftSense requests one of the seven intentions and an optional intended duration. Aggregate session data and 10-second windows are then stored locally while the person browses normally. At the session boundary, the participant reports whether the visit matched the declared purpose. No three-minute risk estimate or reflective prompt is shown in this phase.

Participants use the collector for 10 consecutive study days. The research team checks adherence and technical failure only; it cannot inspect page content because that content is never collected. At the end of the phase, participants transfer the primary and activity-window CSV files through the consented procedure. Files are validated and combined outside the repository.

5.5 Model Development and Freeze

Phase 1 days 1–7 form the development set, while days 8–10 are reserved as a chronological holdout for later sessions from known users. Preprocessing, features, hyperparameters, and threshold rules are selected from days 1–7, using participant-grouped folds within that set. Days 8–10 are evaluated once. A separate participant-grouped or leave-one-participant-out analysis estimates transfer to an unseen user and is not conflated with the chronological result.

The primary candidate is intention-plus-activity logistic regression. It is compared with the time, domain, intention-only, and activity-only baselines, with Random Forest optional as an offline robustness model. Accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices are reported at one, three, and five minutes and at full session. Only information available by a given cutoff may be used at that cutoff.

One three-minute model, its preprocessing parameters, decision threshold, and maximum prompt rate are frozen after evaluation. The threshold is chosen from development data to balance sensitivity, false prompts, and expected burden, and it is not retuned from Phase 2 outcomes. If the model is near chance or produces an impractical eligibility rate, Phase 2 may still serve as a technical feasibility test, but the paper will not claim successful prediction.

5.6 Phase 2: Seven-Day Micro-Randomized Intervention

The same participants use the intervention version for seven study days. Intention collection, sensing, session boundaries, and post-session reflection remain unchanged. A session is eligible only if it is active after three minutes, supplies the required frozen-model features, crosses the threshold, and is not suppressed by the daily cap. Eligible sessions are assigned locally with equal probability:

- (1) **Reflective prompt:** show *Still here for your original reason?* with *Continue intentionally* and *Finish now*.
- (2) **Silent control:** show no mid-session interface.

Assignment is written to the audit log before rendering. Analysis follows the assigned condition even if the participant ignores the prompt. Burden is limited to one prompt per session and no more than three displayed prompts per day. Sessions below threshold, shorter than three minutes, missing required features, or suppressed by the cap remain available for descriptive reporting but are excluded from the randomized effect comparison. Both assignments end with the same reflection question.

5.7 Measures

The primary prediction measures are F1-score and ROC-AUC for the combined model relative to the time and domain baselines. Accuracy, precision, recall, confusion matrices, class balance, calibration where feasible, and the expected intervention rate at the threshold are also reported. Chronological same-participant and participant-held-out results remain separate.

The primary intervention outcome is the proportion of complete binary-labeled eligible sessions reported as drift under each randomized assignment. Secondary measures include total session duration, duration after assignment, immediate session ending, prompt response and frequency, and label completion. Overall Phase 1–Phase 2 differences are descriptive only. At exit, participants rate prompt timing, usefulness for reconsidering intention, annoyance, ease of use, and willingness to continue, and answer one open question about wording or timing. These items assess burden and acceptability, not clinical benefit or objective attention.

5.8 Data Processing and Quality Checks

Preprocessing verifies the participant CSV schema and filename identifier, rejects duplicate identifiers and overlapping sessions, and reports incomplete or non-binary outcomes. Source exports are never modified. Only explicit matched (0) and drifted (1) responses enter the binary analyses; missing and non-binary outcomes stay separate.

Session, activity-window, and intervention records are joined by `session_id`. Logistic regression uses one-hot encoded categories and numeric features standardized from training data only. The fitted transformations are applied unchanged to validation and test partitions. Quality reporting covers participant completion, sessions per participant, class balance, label completion, rejected files, missing windows, intervention eligibility, assignment balance, cap suppression, and attrition.

5.9 Analysis Plan

For RQ1, performance is summarized across the predefined splits, with participant-resampled 95% confidence intervals where feasible. The principal comparison asks whether the combined logistic model improves F1-score and ROC-AUC over time and domain baselines. Intention-only and activity-only results show what each feature group adds. Claims about new-user generalization require support from participant-held-out performance.

For RQ2, the primary analysis includes randomized eligible sessions with complete binary outcomes and follows assigned condition. For participants represented in both assignments, drift proportions are computed separately for prompt and silent sessions. The prespecified simple analysis is a paired Wilcoxon signed-rank test on participant-level proportions, accompanied by the paired difference and a participant-bootstrap 95% confidence interval. Missing-label rates are shown for each assignment. If eligible-session counts permit, a mixed-effects logistic regression with a participant random intercept and prompt assignment as the main fixed effect is added as a sensitivity analysis.

Secondary outcomes are descriptive or use paired participant-level comparisons. Open responses are summarized using a transparent lightweight content analysis. A Phase 1–Phase 2 before–after difference alone will not be presented as proof that an intervention works. Findings will be described as preliminary and short term. No DriftSense performance values, significance tests, or effect sizes will be added until the real consented data have been analyzed.

5.10 Reproducibility and Reporting Commitments

Before recruitment, the protocol will fix session boundaries, intention and reflection wording, label mapping, the prompt cap, exclusions, and the confirmatory/exploratory distinction. Before Phase 2, the build will record a versioned model artifact, preprocessing specification, threshold, and randomization procedure. Participant and session flow will account for short sessions, missing labels or windows, threshold eligibility, cap suppression, assignment, and prompt delivery failure.

Code, schemas, isolated non-participant test fixtures, and analysis scripts may be released after review. Raw participant exports will not be public. A de-identified derived dataset will be considered only when consent, ethics approval, and disclosure-risk review allow it. Otherwise, reproducibility will rely on schema documentation, deterministic preprocessing, fixed seeds, environment details, and disclosure-safe outputs.

5.11 Anticipated Validity Boundaries

The target is the participant's judgment after a session, not an objective measure of attention. Labels may differ between people and may be influenced by memory, social desirability, repeated questions, or the intervention. Label completion and non-binary responses therefore form part of the validity assessment and will be reported by condition.

The RQ2 estimate applies only to sessions that satisfy the frozen eligibility rule and are not removed by the cap; it does not estimate the effect of prompting every session or every user. Randomization supports the prompt-versus-silent comparison within that eligible set, but the fixed order prevents a causal interpretation of the overall difference between phases. The convenience sample of 15–20 people also limits precision and generalization, especially for participant-held-out prediction. Finally, seven days can show short-term feasibility and, at most, preliminary efficacy. It cannot show that any change persists after the extension is removed.

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